

Vaibhavi Agrawal

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LinkedIn | Leetcode | Github

Summary

I'm a motivated and results-driven pre-final year student with a strong interest in machine learning, software development, front-end technologies, management and cybersecurity. Eager to apply technical knowledge and leadership skills to real-world projects, contributing to secure, efficient, and innovative solutions. I'm Dedicated to continuous learning, adaptability, and growth in a collaborative, fast-paced environment.

Technical Skills

- **Programming Languages:** Python, C++, Java, Javascript,HTML, CSS, MySQL
- **Software Development Tools:** VS Code, Git, Google Colab, Postman, Docker, Jupyter Notebook
- **Operating Systems:** Linux (Ubuntu), Windows
- **Frameworks/Technologies:** React, Flask, SQLite, TensorFlow, PyTorch, OWASP, REST APIs
- **Expertise:** Machine Learning, Frontend Development, Web Development, Software Development, Project and Event Management

Education

Degree/Level	Year	Institution/Performance
B. Tech (CSE) Core	2023-Ongoing	Vellore Institute of Technology, Bhopal (CGPA: 9.1/10)
Class XII	2023	GVN The Global School, Bhopal, Madhya Pradesh (85.2%)
Class X	2021	Carmel Convent Senior Secondary School, Bhopal, Madhya Pradesh (96%)

Experience

1. **Media and Content Lead of SDC**
(50K+ Audience Reach, 1M+ Impressions)
2. **PR and Outreach Core team member at ICreate club**
(Handled 500+ Outreach Contacts, Secured 20+ Partnerships and Collabs)
3. **Event Management Core team Member at WiCyS VIT Student Chapter**
(Managed 10+ Events, 1.5K+ Participants, 95 percent Satisfaction Rating)
4. **Community Member at KRAAFT entertainment**
(Active Contributor, I help the team bring our ideas to frame)

Awards and Recognition

Rank	Hackathon	Date
1st	CyberSecureX CTF by KIET	08/06/2025
3rd	IIIT Naya Raipur CTF	06/04/2025
1st	NEC Techfest CTF	15/03/2025
Top 10	GDSC University Development Hackathon	16/08/2025

Projects

1. Improving Object Detection Efficiency through Joint Information Theory and Network Coding Techniques.

- Conducted research on "Improving Object Detection Efficiency through Joint Information Theory and Network Coding Techniques."
- Explored innovative methods to enhance object detection accuracy and computational efficiency.
- Applied network coding principles to optimize data transmission and reduce redundancy.

2. Performance Benchmarking of Conventional and Ensemble Learning Models in Early Lung Cancer Detection

- The research compares traditional machine learning models with ensemble techniques for early lung cancer prediction using clinical and imaging datasets.
- It evaluates accuracy, precision, recall, F-1 Score and AUC-ROC to identify the most reliable diagnostic model.
- Results show that ensemble models outperform conventional approaches, enhancing early detection efficiency and diagnostic reliability.

3. Real-Time Data Optimization for Object Detection Using Information Theoretic Network Coding

- Conducted research on "Real-Time Data Optimization for Object Detection Using Information Theoretic Network Coding."
- It reduces redundant data transmission by optimizing data flow between edge and cloud systems.
- Experimental results demonstrate improved bandwidth efficiency and faster model inferences in dynamic, resource-constrained IoT environments.

4. Shramsetu - Employment Platform for Rural Workforce

- LinkedIn-like employment platform for rural and daily wage workers to connect with verified employers
- Built during GDSC Hackathon and ranked Top 10 out of 150+ participating teams

5. WeSecure Android Application: An app which aims to resolve safety issues through quick alert systems

- Developed a native Android application WeSecure using Java to provide users with a critical, quick-response safety and alert system.
- Implemented a streamlined, one-tap alert feature that immediately broadcasts the user's real-time GPS location and a customized emergency message to pre-configured contacts.
- Utilized Android's Location Services and SMS/Messaging APIs to ensure robust, immediate communication, significantly reducing response time in emergency situations.

Certifications

- NPTEL Fundamentals of Machine Learning: **Score: 72/100**
- FreeCodeCamp: **Responsive Web Design**
- IntelliPaat: **AWS Certification Course**
- AWS Technical Essentials
- Hackerrank 5 star: **Problem Solving in Python**
- Introduction to Matlab: Basic and Features
- Introduction to Problem Solving [Vityrathi]
- Fundamentals in AI/ML [Vityarthi]
- Hackerrank 5 star: **Python**

Co-Curricular Activities

- **Cricket:** Represented in Aarambh 2023 inter-college league (VIT Bhopal).
- **Badminton:** Represented in Aarambh 2024 inter-college league (VIT Bhopal).

Hobbies and Languages

- **Hobbies:** Swimming, Skating, Kho-Kho, Pianist, Nature explorer, Photography
- **Languages:** English, Hindi, French, Japanese(learning)